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ARTICLE

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Psychedelics produce enduring behavioral effects and functional plasticity through mechanisms independent of structural plasticity

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Activation of serotonin 2A ($5\text{-}HT_{2A}$) receptors is thought to underly the long-lasting antidepressant effects of psychedelics such as psilocybin, but beyond that, the molecular and cellular mechanisms involved are not well understood. Recent preclinical studies using mice have primarily examined relatively short time points after psychedelic administration, which does not address the long-lasting effects of psilocybin in humans (i.e., several months or more). We utilized a rat experimental system to demonstrate that both psilocybin and the selective $5\text{-}HT_{2A}$ receptor agonist 25CN-NBOH reduce immobility in the forced swim test without a decrease in effect size for at least three months after a single administration of the psychedelic. There were no overt behavioral differences between psilocybin and 25CN-NBOH treated animals, suggesting $5\text{-}HT_{2A}$ receptor activation is sufficient to produce long-lasting behavioral changes. Functional cellular plasticity in neurons from the medial prefrontal cortex (mPFC) of these animals was assessed using brain slice electrophysiology. Functional plasticity was evident for both psychedelics several months after treatment, and Layer 5 excitatory pyramidal neurons demonstrated significant changes in resting membrane potential, firing rates, and synaptic excitation. Recorded neurons were examined by microscopy for synaptic density and spine classification, which found no differences between control and psychedelic-treated. Gene expression studies for several presynaptic and postsynaptic markers in the mPFC indicated no differences in expression between groups. Together, our results indicate a single treatment with a psychedelic is sufficient to elicit very long-lasting behavioral and cellular changes through enduring function plasticity rather than structural plasticity.

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INTRODUCTION

Psilocybin, a naturally occurring tryptamine found in various mushroom species, has gained wide attention due to its potential to induce long-lasting antidepressant effects in humans. Patients in clinical studies report relief from symptoms for periods ranging from months to years after receiving only one or two doses of psilocybin [1–4]. While psilocybin is advancing towards FDA approval, the precise mechanism underlying its therapeutic effectiveness remains an enigma, especially regarding a single treatment having such long-lasting effects that persist for a year or more in some cases.

On a cellular level, psychedelics can affect changes in gene expression and promote the synthesis of proteins related to synaptic density and neuroplasticity across diverse animal models. This effect has been observed in rats [5, 6], mice [7–9], and pigs [10] following a single administration. Notably, some of these studies showed increases in dendritic spine formation within the prefrontal cortex one day after the administration of psilocybin [9], DOI [7], DMT, and LSD [8]. Few preclinical studies have examined the long-term effects (i.e., several months or more) of psychedelics. Shao et al. [9] found an increase in cortical synaptic density that was detectable for up to 34 days, but only in female mice. Although they examined different regions of the cortex, they did not show changes in spines specific to the mPFC. An open question is whether the structural plasticity observed in the near term (i.e., days to weeks) after treatment persists long term (i.e., several months). Another question is whether $5\text{-}HT_{2A}$ receptor activation is necessary and/or sufficient for these long-term effects because psilocybin activates multiple serotonin receptor subtypes. To address these, we utilized the Wistar Kyoto rat (WKY), a rodent experimental system often used to study depressive-like behaviors [11], to investigate the long-lasting effects of psilocybin, a tryptamine psychedelic, as well as the selective $5\text{-}HT_{2A}$ receptor agonist and phenethylamine psychedelic 25CN-NBOH. For the purpose of this study, we are using the term "psychedelic" as a descriptor for a class of drug that produces effects through activation of $5\text{-}HT_{2A}$ receptors [12]. In these studies, we examined the duration of persistent behavioral effects using the forced swim test (FST), examined expression of genes encoding synaptic structural proteins, performed electrophysiology experiments to

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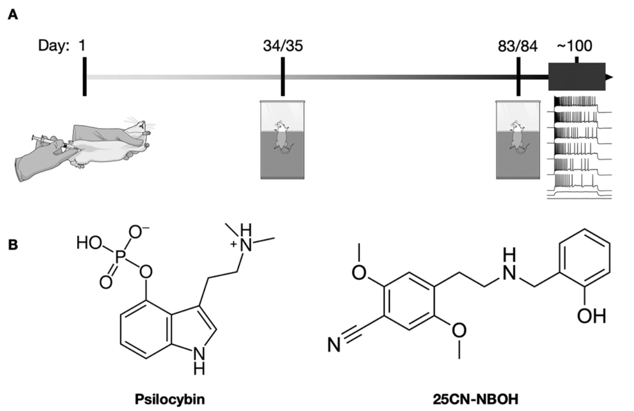

Fig. 1 Experimental design. A Rats were injected with saline vehicle, psilocybin, or 25CN-NBOH on Experimental Day 1. Forced swim tests were performed on days 35 and 84 after administration of psychedelic drugs. Slice electrophysiology was performed within a 2-week period, on average, 100 days post administration. **B** Chemical structures for psilocybin and 25CN-NBOH.

examine enduring functional plasticity, and microscopy to count spine density to inform on enduring structural plasticity in mPFC. We also employed electrophysiology experiments that targeted the infralimbic region of the mPFC due to evidence in the literature noting that activation of the infralimbic-mPFC, and not prelimbic-mPFC, mediates antidepressant-like activity in the FST [13].

METHODS

Animals

Male WKY aged 50–55 postnatal days for each of three treatment conditions were ordered from Charles River Laboratories and randomly assigned to one of three treatment groups, with 8 animals per experimental group ($n = 8$/group). Female rats were excluded from this study, as there is insufficient literature describing their relevant behavioral characteristics to establish that they can be used as an innate model for depression. Rats were pair-housed with a standard 12:12 light-dark cycle, given ad libitum access to water and food and included Bed-r'Nest for enrichment (The Andersons, Inc., Maumee, Ohio). All protocols were approved by the Institutional Animal Care and Use Committee of the LSU Health Sciences Center and were consistent with the Guide for the Care and Use of Laboratory Animals, 8TH edition.

Drugs

Psilocybin was purchased from Cayman Chemical. 25CN-NBOH was synthesized by Jesper Kristensen (University of Copenhagen, Denmark). Each drug (Structures are shown in Fig. 1B) was dissolved in sterile saline and administered in a volume of 1 mL/kg intraperitoneally to each rat corresponding to their assigned treatment group (day 0) (PSI = 1.0 mg/kg; 25CN-NBOH = 1.5 mg/kg). Dosage and route of administration for psilocybin were chosen using guidance from the literature [14] and previous published work from our laboratory [15, 16]. Dosage and route of administration of 25CN-NBOH were chosen based on previous work [17] and comparable at the behavioral level to the dose of psilocybin were used.

Forced swim test

Individuals with major depressive disorder characteristically use passive or avoidant coping strategies when challenged with adversity [18–22], as do stress-vulnerable rats used as models for human depression pathology [23–25]. The FST is a rodent paradigm that interprets passive coping strategy (immobility) as a behavioral metric comparable with that characteristic of depressed humans. The FST measures coping strategies that are not exclusively mediated by neurocircuits known to be dysfunctional in humans and thus lacks construct validity. However, the FST remains the most reliable measure of passive coping strategy in both male and female rats [26] and can be used to screen for antidepressant-like action of drugs given to otherwise healthy rats [27, 28] for passive coping strategies in animals modeling depression pathology [29] and for antidepressant-like action of drugs given to animals modeling depression pathology [29, 30]. The FST has most frequently been used to screen drugs for potential antidepressant-like effects with a subacute dosing strategy in which the drug is administered three times prior to testing—1, 5, and 23.5 h prior to testing [28]. When using the FST to evaluate the effects of a drug administered within 24 h of testing, rats that have had prior interventions such as surgeries or stressors need their FST behaviors evaluated both before and after the drug is given to ensure that the effects are attributable to the drug, rather than to the prior intervention [28]. However, when no drug is given in the day between pre-exposure and testing, evaluating the rats' behaviors twice is redundant and not informative. To control for potential nonspecific sedative or stimulant effects of the test drug, overall LCA is assessed immediately prior to the FST [28]. Increased immobility in the FST is only meaningful if overall LCA is unchanged.

A 15-min pre-exposure to the paradigm one day prior to testing is used to elicit consistent passive coping strategy (immobility) in otherwise healthy rats [28]. During pre-exposure, rats were placed into a plastic cylindrical tank (114 cm × 30.5 cm) that contained 30 cm of water at 25–28 °C. The water depth was such that the rats could not support themselves by touching the bottom of the tank with their hind paws, and their tails could not touch the bottom of the tank while keeping their noses above water. After a 15-min swim, the rats were removed, dried with paper towels, and returned to their home cages. The pre-exposure was not recorded. Fresh water was used for each animal. During the testing session the following day, a video camera was mounted to the side of the tank and the rats were exposed to a 5-min swim under the conditions described above. Immediately after testing, animals were removed, dried with paper towels, and returned to their home cages. The 5-min swim was recorded for later scoring for immobility, swimming, and climbing. FST scoring was performed by trained scorers blind to the treatment conditions and employed the modified sampling technique [28]. Immobility was defined as no active attempts to escape while maintaining a floating posture in which the rats made only the movements necessary to keep their heads above water. Swimming was defined as active attempts to escape with movements directed laterally towards the sides of the tank. Climbing was defined as active attempts to escape with movements directed vertically towards the upper edge of the tank. High immobility in control (SAL) WKY rats indicates increased passive coping, a behavior characteristic of people with major depressive disorder; rats with significantly less immobility than the control (SAL) group indicate an antidepressant-like effect of the drug.

Locomotor activity

Locomotor activity (LCA) was assessed immediately prior to each FST to prevent confounding by nonspecific sedative or stimulant effects. LCA area consists of an open field square arena (61 cm^2) with 45 cm high opaque walls. Assessment was performed by placing each rat into the arena and allowing them to explore freely for 5 min. The rats' movements within the arena were recorded and later scored for distance traveled (cm) using EthoVision XT 8.5 tracking software (Noldus Information Technology).

Statistical analyses and exclusions for behavioral experiments

Immobility in the FST was analyzed with repeated measures analysis of variance (ANOVA) with Holm-Sidak *post hoc* for the independent variable of *treatment* (SAL, PSI, 25 CN-NBOH) at *times* (5- and 12-weeks post-injection). Equipment failure resulted in the exclusion of four videos of LCA from Week 5—one from the control group, one from the PSI group, and two from the 25CN-NBOH group. Because of excluded data points, LCA was compared with mixed-effects analysis restricted maximum likelihood (REML) of distance traveled using the independent variable *treatment* (SAL, PSI, 25C-NBOMe) at *times* (5 and 12 weeks). Behavioral data are expressed as mean ± standard error of the mean, and were analyzed using Prism 10 (GraphPad Software, Boston, MA).

Brain electrophysiology

In preparation for electrophysiological experiments, animals were intracardially perfused with an N-methyl-D-glucamine (NMDG) based artificial cerebrospinal fluid (aCSF) under deep isoflurane anesthesia. Brains were removed and immediately hemisected. One half was flash frozen on dry ice, and the other was used for electrophysiology and sliced in room temperature aCSF containing (in mM): 92 NMDG, 2.5 KCl, 1.25 NaH_2PO_4, 30

NaHCO$_3$, 20 HEPES, 25 glucose, 2 thiourea, 5 Na-ascorbate, 3 Na-pyruvate, 0.5 CaCl$_2$, 10 MgSO$_4$; pH is titrated to 7.3–7.4 with concentrated hydrochloric acid [31]. Coronal slices containing infralimbic-mPFC were then transferred to a holding chamber (containing the same NMDG solution as slicing) in a water heat bath kept at 37 °C. After 12 min, slices were removed from the holding chamber in the heat bath and allowed to rest at room temperature in a chamber containing holding solution for 45 min before electrophysiological recordings. The holding solution consisted of (in mM): 92 NaCl, 2.5 KCl, 1.25 NaH$_2$PO$_4$, 30 NaHCO$_3$, 20 HEPES, 25 glucose, 2 thiourea, 5 Na-ascorbate, 3 Na-pyruvate, 2 CaCl$_2$, 2 MgSO$_4$.

Whole-cell recordings

At the time of recording, slices were transferred to the slice chamber in the electrophysiology rig, in a recording aCSF kept at 30–32 °C by an in-line heater. The recording solution consisted of (in mM): 119 NaCl, 2.5 KCl, 1.25 NaH$_2$PO$_4$, 24 NaHCO$_3$, 12.5 glucose, 2 CaCl$_2$, 2 MgSO$_4$. Whole-cell recordings were performed using borosilicate glass micropipettes (3–7 MOhm) filled with internal solution containing (in mM): 130 K-gluconate, 10 HEPES, 10 Na$_2$-phosphocreatine, 4 MgCl$_2$, 4 Na$_2$-ATP, 0.4 Na-GTP, 3 ascorbic acid, 0.2 EGTA (pH 7.25, 290–295 mOsm). All pipettes also contained 0.2% biocytin to enable *post hoc* visualization of recorded neurons using streptavidin-conjugated fluorophores. Recordings were obtained from Layer 5 pyramidal neurons in the infralimbic-mPFC [32].

Different current clamp and voltage clamp recordings were carried out and used to analyze different measures of intrinsic neural excitability and synaptic properties. The details are provided in our previous work [33]. Briefly, we estimated resting membrane potential (RMP), input resistance, firing rate rheobase (i.e., minimum injected current to produce action potentials), suprathreshold action potential firing rate, and spontaneous excitatory postsynaptic current (sEPSC) rate and amplitude.

Electrophysiology group sizes

8 animals per group (saline, psilocybin, 25CN-NBOH) were used for a total of 24 animals. The neural recording numbers per experimental group and firing phenotype subgroupings are as follows: saline $n = 37$ (11 intrinsic bursters [IB]; 13 rhythm oscillatory bursters [ROB]; 13 regular spike [RS]), psilocybin $n = 38$ (11 IB; 10 ROB; 17 RS), and 25CN-NBOH $n = 34$ (17 IB; 13 ROB; 4 RS).

Data analysis for electrophysiology experiments

All numerical analysis and statistics were performed in MATLAB (Mathworks, Natick, MA). To assess overall patterns of functional plasticity, we used principal component analysis (PCA), a statistical method that reduces complex, multidimensional datasets into a smaller number of new variables (principal components) that capture the primary sources of variation across the data [34]. Rather than evaluating each electrophysiological property (e.g., resting membrane potential, input resistance, firing rate, etc.) in isolation, PCA enables us to detect coordinated patterns of change across all parameters, which may reveal group-level effects that are not apparent through univariate analyses alone.

For analysis of individual electrophysiological metrics (e.g., resting membrane potential, input resistance, rheobase, firing rate, sEPSC amplitude and frequency), one-way ANOVAs were conducted separately for adapting and bursting neuron types across treatment groups (SAL, PSI, 25CN-NBOH). When the main effect of treatment was significant, Tukey's HSD post hoc comparisons were used to identify pairwise group differences. Post hoc tests were only performed following a significant main effect of treatment in the corresponding ANOVA.

In our analysis, we included all intrinsic and synaptic parameters from recorded neurons and normalized each parameter to unit variance before performing PCA. This normalization step ensures that all variables contribute equally to the analysis regardless of their original scale or units, so that parameters like membrane potential (measured in mV) do not dominate over others like firing rate or sEPSC frequency. The first principal component (PC1) reflects the direction of greatest variance in the dataset—in our case, primarily influenced by differences in resting membrane potential and input resistance—while the second (PC2) reflects the next-largest source of variability, largely driven by firing rate and rheobase. These composite variables allow us to test whether psychedelic treatment induces a coordinated shift in electrophysiological phenotype, even if no single measure alone shows a strong effect.

Tissue preparation and imaging

After the electrophysiology experiment ceased, brain slices were fixed in 4% paraformaldehyde for 24 h at 4 °C. Using a 24-well plate, slices were washed three times for 10 min each in 1X PBS with 0.01% sodium azide to prevent bacterial and fungal growth. Slices remained in this solution until all electrophysiology experiments were complete, to carry out staining on all slices simultaneously. For staining, slices were washed two times in 1X PBS with 0.3% Triton (PBST) at room temperature for 10 min. Secondary antibody Streptavidin 568 was added to 1X PBS at a 1:1000 concentration for 24 h at 4 °C. The following day, slices were washed for 10 min in PBST at room temperature, followed by two 10-min washes in 1X PBS. Slices were mounted on Superfrost Plus Microscope Slides with Prolong Gold Antifade Reagent. Images were captured using a Leica SP8 confocal microscope (Leica Biosystems, Buffalo Grove, IL) with the following settings: 16-bit images, frame average 2, line average 8, magnification 100×, z-step size 0.06 µm, 200 µm away from the soma, 100 µm dendrite length per image.

Spine counting

Dendritic spine analyses were performed on three-dimensional projections of the confocal images of the dendrites described above from recorded neurons filled with biocytin using ImageJ [35]. Spines were classified into three categories based on their visual appearance: (i) mushroom spines were dendritic extensions with a head diameter >0.5 µm or 2× the spine neck diameter; (ii) stubby spines were dendritic extensions with no discernable head and neck length of ≤0.5 µm; and (iii) thin spines were dendritic extensions with a length of >0.5 µm and a head diameter <0.5 µm or no discernable head. Spines were counted twice and averaged for final counts. In total $n = 13$, 17, 8 neurons were scored from the saline, psilocybin, and 25CN-NBOH treatment groups, with an average of 115 spines scored per neuron across all treatment groups.

Analysis of gene expression

The remaining left hemisphere of each brain was used for qRT-PCR experiments. The left hemisphere was flash-frozen in 2-methyl butane and stored at −20 °C until dissection. The infralimbic-mPFC was dissected out, RNA purified, and expression of several genes encoding for synaptic proteins was assessed using methods cited in Hibicke et al. [16]. Briefly, the left hemisphere was brought to −14 °C and sectioned into 500 µm coronal sections using a Leica CM3050 S cryostat (Leica Biosystems, Buffalo Grove, IL). The infralimbic-mPFC was dissected under an Olympus SZ51 microscope (Olympus Corp., Center Valley, PA) on a Corning CoolRack® M30 (Corning, Inc., Corning, NY) kept between 0 °C and −10 °C. Homogenization was carried out with RNA/Protein Purification Plus Kit (Norgen Biotek Corp., Ontario) according to the manufacturer's directions. cDNA was synthesized from RNA using the ImPromp-II™ kit (Promega Corp., Madison, WI). Primers and probes (Supplementary Table 1) were designed using IDT PrimerQuest™ Tool and synthesized by IDT (Integrated DNA Technologies, Newark, NJ). Duplicate amplification reactions were carried out using Roche 480 LightCycler® II (Roche Molecular Systems, Inc., Indianapolis, IN). The Roche software package calculated the amplification threshold using 2nd derivative maximum analysis. $\Delta\Delta^{ct}$ values were calculated by subtracting the amplification threshold of *ywhaz* from the amplification threshold of the gene of interest. These values were normalized to control animals to calculate percent change. Statistical analysis was performed using GraphPad Prism 10 (GraphPad Software, Boston, MA). One-way ANOVA was performed with Tukey's multiple comparison tests for *post hoc* analysis to compare expression between samples from different treatment groups ($n = 8$/treatment group).

RESULTS

A single dose of a psychedelic elicits persistent, long-lasting antidepressant-like effects

Rats were evaluated in the FST 35 days (5 weeks) and 84 days (12 weeks) after a single administration of drug or vehicle (Fig. 1A). Each treatment group had eight animals per group ($n = 8$). Both *treatment* ($F_{(2,21)} = 20.28$, $p < 0.0001$) and *subject* ($F_{(21,21)} = 3.068$, $p = 0.0066$) were significant sources of variation in immobility, but *time* was not a significant source of variation, and no significant interactions were observed between *treatment* and *time*. Although the behaviors measured were largely quite similar, as noted below,

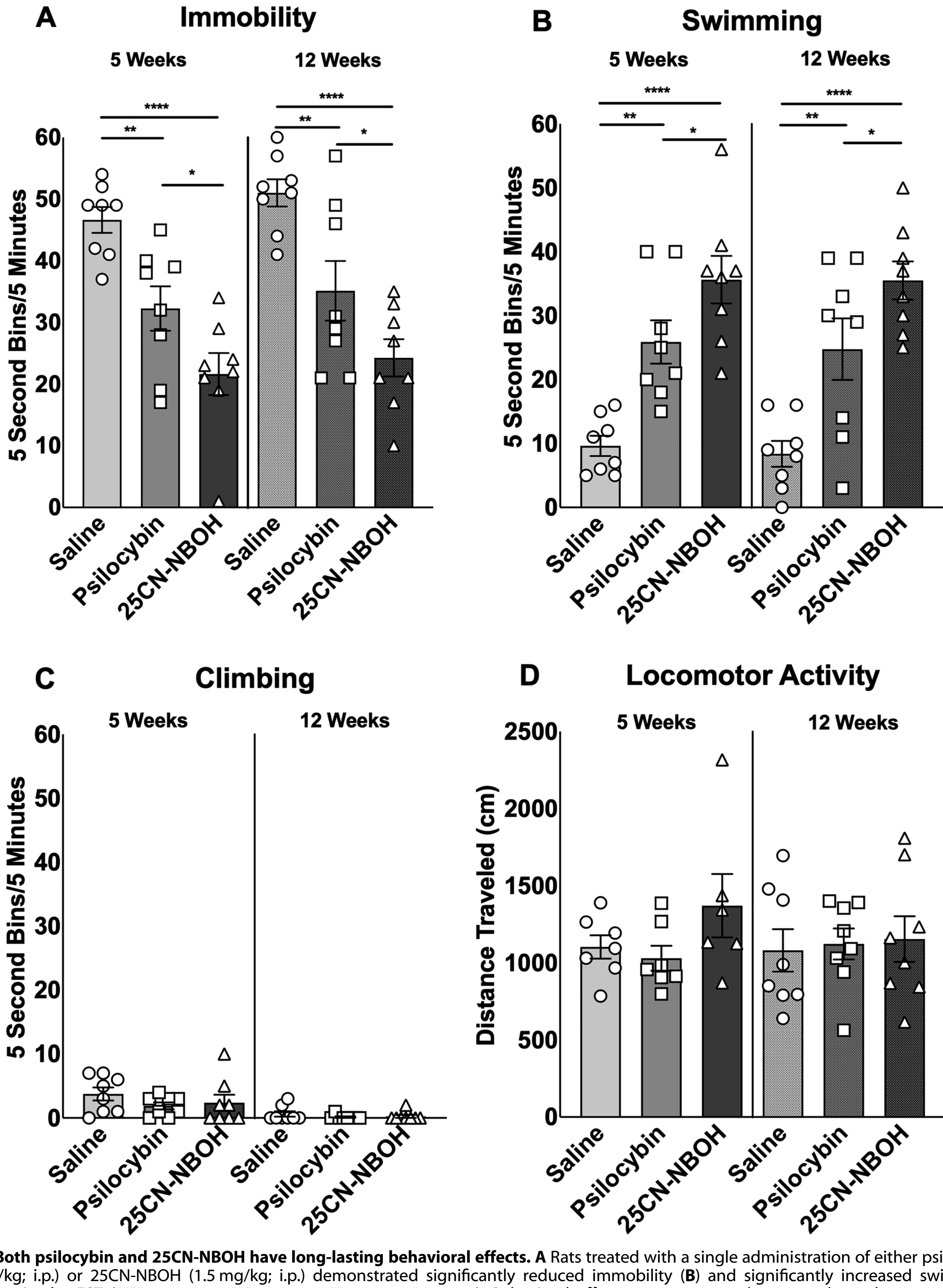

Fig. 2 Both psilocybin and 25CN-NBOH have long-lasting behavioral effects. **A** Rats treated with a single administration of either psilocybin (1.0 mg/kg; i.p.) or 25CN-NBOH (1.5 mg/kg; i.p.) demonstrated significantly reduced immobility (**B**) and significantly increased swimming behaviors in the FST (****$p < 0.0001$, ***$p < 0.001$, **$p < 0.01$, *$p < 0.05$). Behavioral effects were measured at 5 weeks and 12 weeks post administration. No decrease in effect size was measured between weeks 8 and 12. **C** No changes in climbing behaviors were observed among groups. **D** Locomotor activity was also measured immediately prior to FST testing at weeks 5 and 12. No significant differences were observed between treatment groups.

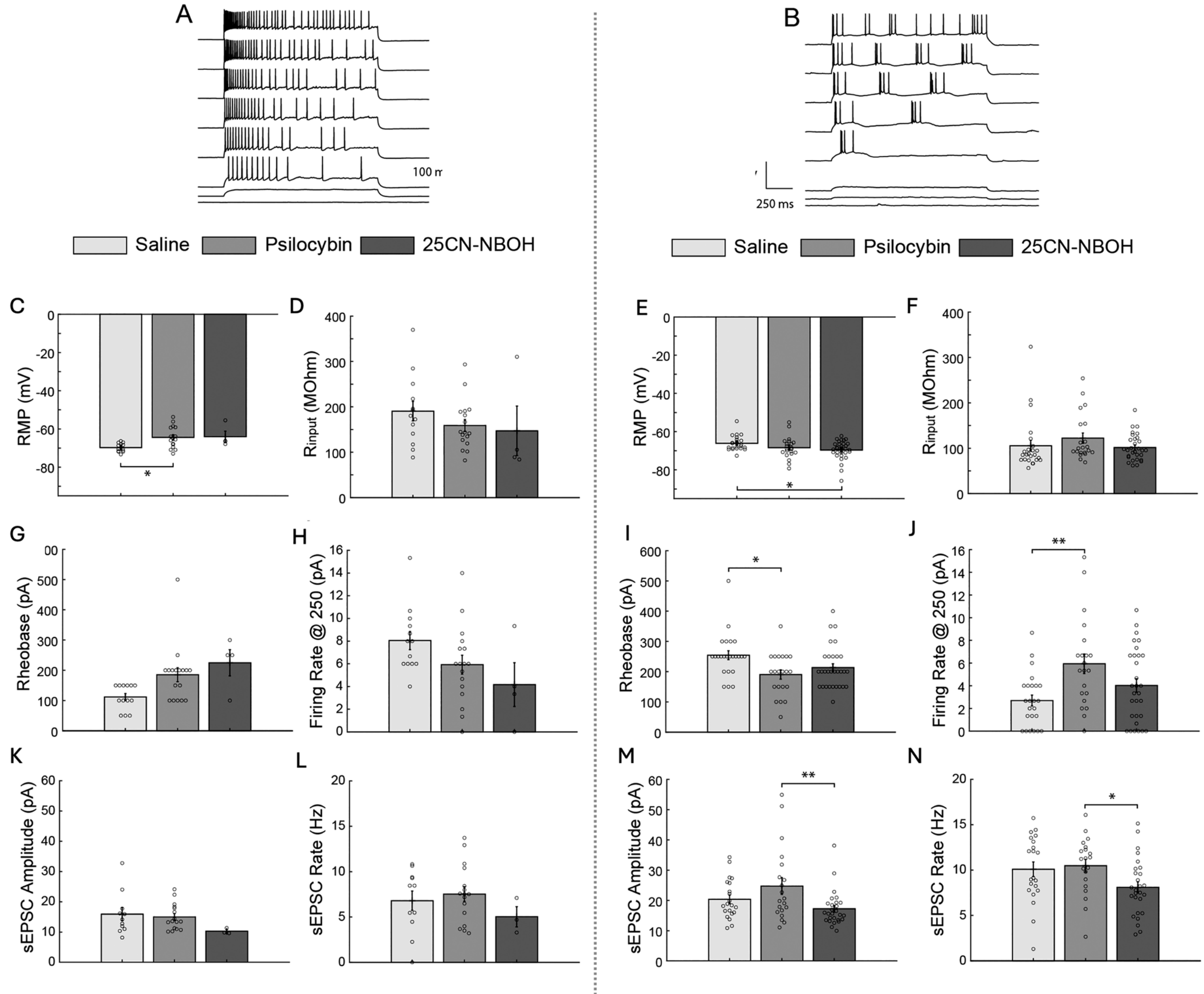

Fig. 3 Cell-type effects of psilocybin and 25CN-NBOH in intrinsic and synaptic properties of L5 infralimbic-mPFC neurons. **A** The membrane voltage in response to a series of depolarizing current injections in an adapting neuron exhibits a temporal firing pattern where there is an initial high action potential frequency response, which gradually adapts to a lower frequency response. **B** The membrane voltage response in an example bursting neuron exhibits a response where a "burst" is produced to injected current, where there is a series of action potentials with very short interspike intervals followed by a long interspike interval. **C** The resting membrane potential in adapting neurons shows a significant effect of psychedelic drug treatment ($F_{(1,32)} = 0.0106$, $p = 0.0106$). *Post hoc* analysis shows significant differences between saline and psilocybin neurons ($p = 0.0126$). **D** There is no effect of treatment on input resistance in adapting neurons. **E** The resting membrane potential in bursting neurons shows a significant effect of psychedelic treatment ($F_{(1,73)} = 3.41$, $p = 0.0384$). *Post hoc* analysis shows significant differences between psilocybin and 25CN-NBOH neurons ($p = 0.0298$). **F** There is no effect of treatment on input resistance in bursting neurons. **G** There is no effect of treatment on rheobase in adapting neurons. **H** There is no effect of treatment on firing rate in adapting neurons. **I** The rheobase in bursting neurons shows a significant effect of psychedelic drug treatment ($F_{(1,74)} = 4.73$, $p = 0.0117$). *Post hoc* analysis shows significant differences between control and psilocybin neurons ($p = 0.0101$). **J** The firing rate in bursting neurons shows a significant effect of psychedelic drug treatment ($F_{(1,74)} = 5.74$, $p = 0.0048$). *Post hoc* analysis shows significant differences between control and psilocybin neurons ($p = 0.0033$). **K** There is no effect of drug treatment on sEPSC amplitude in adapting neurons. **L** There is no effect of treatment on sEPSC rate in adapting neurons. **M** The sEPSC amplitude in bursting neurons shows a significant effect of psychedelic drug treatment ($F_{(1,67)} = 4.58$, $p = 0.0137$). *Post hoc* analysis shows significant differences between psilocybin and 25CN-NBOH neurons ($p = 0.0098$). **N** The sEPSC amplitude in bursting neurons shows a significant effect of psychedelic drug treatment ($F_{(1,67)} = 3.6$, $p = 0.0329$). Further, post hoc analysis shows significant differences between psilocybin and 25CN-NBOH neurons ($p = 0.0455$).

there were some differences between the psilocybin-treated and 25CN-NBOH-treated groups. Rats given PSI were significantly less immobile than control rats at 5 weeks ($p = 0.0077$) and 12 weeks ($p = 0.0032$) after administration (Fig. 2A). Rats given 25CN-NBOH were significantly less immobile than control rats at 5 weeks ($p < 0.0001$) and 12 weeks ($p < 0.0001$) after administration (Fig. 2A), and less immobile than rats given PSI at 5 weeks ($p = 0.0290$) and 12 weeks ($p = 0.0257$). Similarly, both *treatment* ($F_{(2,21)} = 22.92$, $p < 0.0001$) and *subject* ($F_{(21,21)} = 2.753$, $p = 0.0123$) were significant sources of variation in swimming, but *time* was not a significant source of variation. Rats given PSI swam significantly more than control rats at 5 weeks ($p = 0.0021$) and 12 weeks ($p = 0.0020$) after administration (Fig. 2B). Rats given 25CN-NBOH swam significantly more than control rats at 5 weeks ($p < 0.0001$) and 12 weeks ($p < 0.0001$) after administration (Fig. 2B), and more than rats given PSI at 5 weeks ($p = 0.0408$) and 12 weeks ($p = 0.0249$). No differences were observed within individual treatment groups between 5- and 12-week time points, indicating no loss of effect size

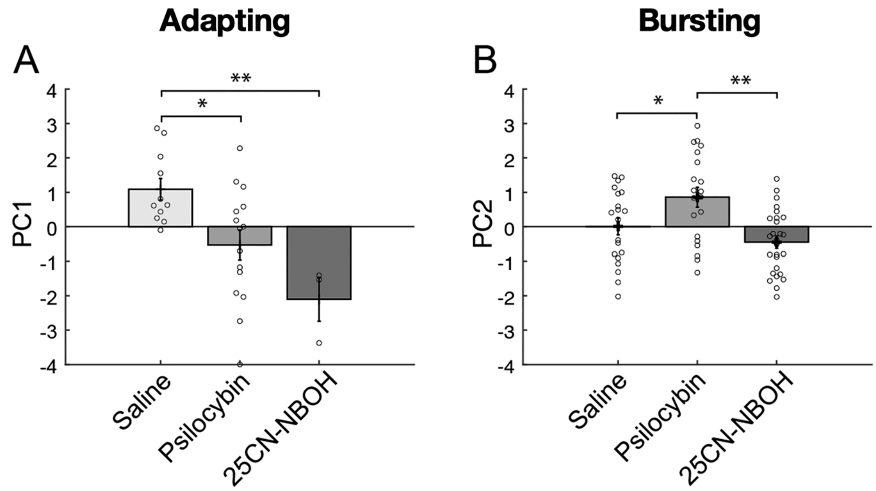

Fig. 4 Neural property principal component analysis. A The principal component 1 (PC1) in adapting neurons shows a significant effect of psychedelic drug treatment ($F_{(1,28)} = 7.47$, $p = 0.0027$). *Post hoc* analysis shows significant differences between control and psilocybin neurons ($p = 0.0216$) and control and 25CN-NBOH ($p = 0.0054$). **B** The principal component 2 (PC2) in adapting neurons shows a significant effect of psychedelic drug treatment ($F_{(1,66)} = 8.39$, $p = 0.0006$). *Post hoc* analysis shows significant differences between control and psilocybin neurons ($p = 0.0422$) and psilocybin and 25CN-NBOH ($p = 0.0004$).

for each treatment over time (Fig. 2A, B). No differences were observed in climbing behaviors (Fig. 2C), or in overall LCA (Fig. 2D).

Psilocybin and 25CN-NBOH produce long-lasting functional plasticity in the mPFC

Approximately 100 days post drug administration, brains were harvested, and neurons in the infralimbic region of the mPFC layer 5 were recorded from (Fig. 1A). Neurons were classified and subgrouped according to the temporal characteristic of their action potential firing patterns. In adapting neurons, the membrane voltage in response to a series of depolarizing current injections exhibits a temporal firing pattern where there is an initial high action potential frequency response which gradually adapts to a lower frequency response (Fig. 3A). The membrane voltage response in an example bursting neuron exhibits a response where a “burst” is produced in response to injected current where a series of action potentials with very short interspike interval followed by a long interspike interval (Fig. 3B). Adapting-neurons putatively project intratelencephalically while bursting neurons putatively project to subcortical target, excluding the thalamus [36].

In adapting neurons, we found a significant main effect of psychedelic drug treatment on resting membrane potential (RMP; $F_{(1,32)} = 6.97$, $p = 0.0106$). *Post hoc* analysis revealed that psilocybin-treated neurons were significantly more depolarized than saline controls ($p = 0.0126$; Fig. 3C). No significant treatment effects were observed on input resistance ($F_{(1,32)}$ not significant; Fig. 3D), rheobase ($F_{(1,32)}$ not significant; Fig. 3G), or firing rate ($F_{(1,32)}$ not significant; Fig. 3H). Similarly, sEPSC amplitude and frequency were not significantly affected by treatment in adapting neurons ($F_{(1,32)}$ not significant; Fig. 3K, L).

Notably, only four regular spiking (RS) neurons were recorded from the 25CN-NBOH group, limiting our ability to evaluate treatment-related effects in this subclass. While we cannot rule out that this low count could reflect a treatment-related shift in cell type representation, it also may have occurred by chance due to random variation in cell sampling. Additional recordings will be needed to clarify whether the observed distribution reflects a true biological effect.

In contrast, bursting neurons exhibited broader functional changes. RMP showed a significant treatment effect ($F_{(1,73)} = 3.41$, $p = 0.0384$), with psilocybin-treated neurons differing significantly from the 25CN-NBOH group ($p = 0.0298$; Fig. 3E). Input resistance did not differ significantly between treatment groups ($F_{(1,73)}$ not significant; Fig. 3F). Rheobase was significantly reduced by treatment ($F_{(1,74)} = 4.73$, $p = 0.0117$), with psilocybin-treated neurons showing lower rheobase compared to controls ($p = 0.0101$; Fig. 3I). Firing rate also showed a significant main effect of treatment ($F_{(1,74)} = 5.74$, $p = 0.0048$), with psilocybin-treated neurons firing more rapidly than controls ($p = 0.0033$; Fig. 3J). For synaptic properties, bursting neurons showed a significant effect of treatment on sEPSC amplitude ($F_{(1,67)} = 4.58$, $p = 0.0137$), with psilocybin-treated neurons exhibiting higher amplitudes than the 25CN-NBOH group ($p = 0.0098$; Fig. 3M). sEPSC frequency was also affected ($F_{(1,67)} = 3.6$, $p = 0.0329$), with a significant difference between psilocybin and 25CN-NBOH-treated neurons ($p = 0.0455$; Fig. 3N).

To assess whether psychedelic treatment produced coordinated shifts across intrinsic and synaptic properties, we conducted PCA on the full dataset. In adapting neurons, **PC1**—driven primarily by variance in RMP and input resistance—was significantly reduced in both psilocybin and 25CN-NBOH groups compared to controls ($F_{(1,28)} = 7.47$, $p = 0.0027$; psilocybin vs. control $p = 0.0216$, 25CN-NBOH vs. control $p = 0.0054$; Fig. 4A), suggesting similar shifts in passive physiological properties.

In bursting neurons, **PC2** - driven largely by rheobase and firing rate - was significantly increased in the psilocybin group ($F_{(1,66)} = 8.39$, $p = 0.0006$), with *post hoc* tests indicating a significant increase relative to both saline ($p = 0.0422$) and 25CN-NBOH ($p = 0.0004$; Fig. 4B). These PCA findings support the conclusion that psilocybin produces a distinct pattern of enhanced excitability in bursting neurons, not observed with 25CN-NBOH. Overall, the PCA results highlight coordinated, cell type-specific plasticity patterns that are not fully captured by univariate comparisons.

5-HT_{2A} receptor agonists do not increase spine density or levels of transcripts for synaptic structural genes 12 weeks post-treatment

Neurons from electrophysiology experiments were backfilled with biocytin, stained with streptavidin, and imaged by confocal microscopy to assess synaptic density and spine morphology in recorded cells (Fig. 5B). We found no differences in synaptic density or number of spine-type due to psychedelic drug administration (Fig. 5B). Additionally, we found no differences in expression of several genes tested encoding for several synaptic

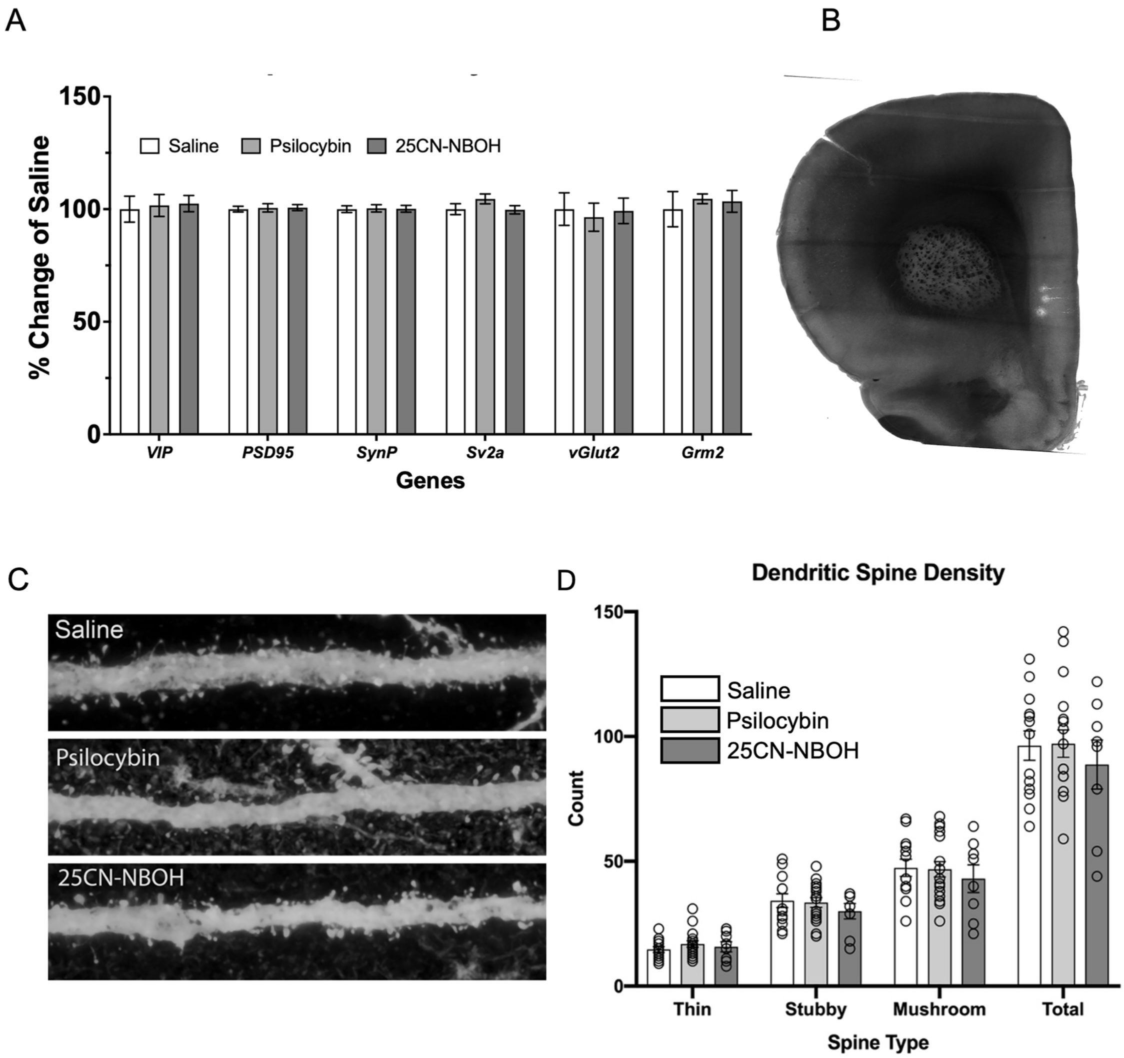

Fig. 5 Synaptic spine density was not different between treatment groups when measured after 12 weeks. A The mRNA expression of several genes encoding for proteins related to synaptic structure and function was tested by qRT-PCR. No significant differences were found between any of the treatment groups. Error bars represent standard error of the mean. **B** Example coronal brain slice showing neurons recorded from filled with biocytin. **C** Representative confocal images of filled and counted neurons within a slice for each treatment group. **D** Counts of each spine type within the 100 micron region of the proximal dendrites (inset: representative high resolution dendrite image). No differences in the number of spine types were measured between treatment groups. No significant differences were found between any of the treatment groups for either total number of individual types of spines. Error bars represent standard error of the mean.

proteins (Fig. 5A). Statistical analysis was performed using GraphPad Prism 10 (GraphPad Software, Boston, MA). One-way ANOVA was performed with Tukey's multiple comparison tests for *post hoc* analysis.

DISCUSSION

Preclinical studies examining potential therapeutic mechanisms for psychedelics to treat psychiatric disorders have largely focused on the acute effects with drug on board or in a window ranging from one hour to 30 days after psychedelic drug administration. These studies have reported that psychedelics can rapidly produce increases in synaptic density and plasticity, with increases in spine density still detectable after several days to several weeks [7–9]. None of these studies, however, has informed us on the mechanisms underlying the persistence of behavioral effects commonly seen in humans that can last several months or more. We previously examined the brains of animals subject to mild restraint stress during adolescence and treated with a single dose of psilocybin one week after the stress protocol ended. We found that there were significant antidepressant-like behavioral effects after five weeks, but no detectable changes in expression of the genes for several synaptic-related proteins [16]. Our previous studies only assessed behavioral outcomes at time points out to 5 weeks after psychedelic drug administration [15, 16]. Here, we further measured FST behavior at 12 weeks (84 days) post psychedelic drug administration. We found significant antidepressant-like effects for immobility in the FST at 5 weeks for both drugs. At 12 weeks, the magnitude of the effect size from week 5 to week 12 for either psilocybin or 25CN-NBOH-treated rats was not statically different, indicating the behavioral effects following a single administration of a psychedelic are indeed long-lasting and persistent (Fig. 2A, B). Interestingly, we did note some behavioral differences in FST behaviors between the two psychedelics. We speculate that this may be due to differential receptor engagement. At the dose of drug we used, 25CN-NBOH is predicted to primarily engage with 5-HT_{2A} receptors, and not significantly to any other serotonin receptor type with the possible exception of 5-HT_{2C} [17, 37]. psilocybin, on the other hand, is likely engaging with nearly all other serotonin receptors [38]. Regarding

5-HT_{2A} receptors, some reports have questioned the role of this receptor in the antidepressant-like effects mediated by psychedelics [39]. Our results with 25CN-NBOH support the notion that 5-HT_{2A} receptor activity alone is sufficient to elicit long-lasting changes in behavior and synaptic function. Our results do not, however, preclude the involvement of other serotonin receptors in the overall effects of psilocybin, and need further validation incorporating a receptor-selective antagonist at relevant doses.

We focused our attention on the infralimbic-mPFC for our molecular, structural, and electrophysiological experiments because previous studies have demonstrated that mobility behavior in the FST is mediated by projections from the infralimbic-mPFC to the dorsal Raphe nucleus [13]. Previous in vitro whole-cell recordings in rat mPFC Layer 5 pyramidal neurons have shown that psychedelics (e.g., DOI, serotonin analogues) acutely increase both the amplitude and frequency of spontaneous EPSCs via 5-HT_{2A} receptor activation [40]. However, most electrophysiological data with psychedelics have been collected after acute exposure, and no studies have evaluated the long-term changes several months after in vivo administration. We found that both psilocybin and 25CN-NBOH produced significant changes in the electrophysiological properties of neurons several months after a single administration. While some changes were conserved, some differed between the two psychedelics, and responses were not simple. For example, psilocybin significantly depolarized adapting neurons and increased the excitability of bursting neurons, as evidenced by reduced rheobase and increased firing rate. In contrast, 25CN-NBOH had less pronounced effects on excitability; it did not significantly alter firing rates or rheobase but was associated with a relative hyperpolarization of RMP in bursting neurons compared to psilocybin. These divergent effects on intrinsic properties suggest that psilocybin and 25CN-NBOH engage distinct intracellular mechanisms or receptor targets despite producing similar behavioral outcomes. PCA demonstrated overall effects on two distinct components, with the direction of change from control the same for adapting but opposite for bursting neurons. These differences between psychedelics may be due to differences in receptor binding and activity profiles. Psilocybin is likely pharmacologically active at nearly all serotonin receptors at the dose used, whereas 25CN-NBOH is likely only pharmacologically active at 5-HT_{2A} receptors. Because 5-HT_{1A} and 5-HT_{2C} activity functionally oppose 5-HT_{2A} receptor activity, we speculate that increased effect size for 25CN-NBOH at a behavioral equivalent dose of psilocybin may be due to full activity of 5-HT_{2A} receptors without the inhibition from 5-HT_{1A} and/or 5-HT_{2C} produced by psilocybin. Although we have demonstrated a complex long-lasting functional plasticity, it remains to be determined in future experiments the precise mechanism and role of these possible effects and the individual neuronal cell types identified in behavioral responses.

Because of the well-documented structural plasticity that can be invoked by psychedelics, as demonstrated by several independent reports [7–9], we cannot discount this process in their putative therapeutic effects for psychiatric disorders. Based on our results and those of others, we propose here that psychedelics induce a rapid structural plasticity involving increases in spine density within hours to days after a single administration, with this increased structural plasticity contributing to the observed rapid antidepressant effects. However, we hypothesize that after an early peak of structural growth, perhaps after several weeks, but not months, synapses begin to prune back, and the number of spines and synapses returns to baseline. In the case of ketamine, whose antidepressant-like effects last only about two weeks in both humans [41] and preclinical models [15], the pruning could simply result in a recapitulation of the baseline depressed state. Work presented by Phoumthipphavong et al. [42] supports this hypothesis and shows that ketamine induces a rapid increase in dendritic spine formation peaking at 24 h post ketamine injection, followed by a gradual decrease that is accompanied by an increase in spine elimination. At 15 days post-ketamine, they show that spine elimination in the ketamine-treated group is at a higher rate than in the saline-treated group [42]. Interestingly, we found no significant correlations between spine density, performance in the FST, or electrical properties of the recorded neurons in our system (data not shown). These findings suggest that the behavioral and electrophysiological effects of psychedelics involve complexities that cannot be explained by spine density alone. Because very long-lasting changes in biological function often involve epigenetic changes influencing the expression of genes, we speculate that psychedelics like psilocybin and 25CN-NBOH induce epigenetic changes that contribute to the persistent expression of one or more ion channels that alter conductance, leading to more efficiency in the remaining synapses and the observed functional plasticity. We further speculate that persistent functional plasticity underlies the long-lasting antidepressant-like behaviors in preclinical models and long-lasting therapeutic effects in humans.

DATA AVAILABILITY

All data are available upon request.

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AUTHOR CONTRIBUTIONS

HMK designed and carried out experiments, analyzed data, and contributed to writing the manuscript; MH designed and carried out experiments, analyzed data, and edited the manuscript; JM performed the electrophysiological experiments, analyzed data, and edited the manuscript. AJ analyzed the dendritic spine data; JLK contributed to the design and interpretation of data from the behavioral experiments and edited the manuscript; CDN designed experiments, analyzed data, and contributed to the writing of the manuscript.

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COMPETING INTERESTS

CDN is a founder and board member of 2A Biosciences. The other authors declare no competing interests.

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